



Route Study

Presented to the Board of Education
of Brecksville Broadview Heights
City School District
November 20, 2019



Methodology:

In general, costs of operation in a school bus fleet are tied directly to the number of buses operated daily. Typically, the use of more buses results in higher cost, both for capitalization of the vehicle purchase and for total cost of operation.

Efficiency studies completed with other school districts and with districts that have a large geographical area have found it more cost effective for one school bus to make two or more trips to school than it is for two buses to make one trip to school.



Determining Ridership

Performed by routing software that identified:

- 83% of elementary students are bus riders
- 85% of middle school students are bus riders
- 40% of high school students are bus riders.

Impacts if the district continues to transport students residing with-in one mile of the buildings.

- Elementary Ridership could reach 90% of the total ADM.
- Ridership percentage for HS/MS should remain constant at the current level.



Optimal routing requires

- Similarly sized groups of students are schedule at intervals that allow buses to run multiple routes.
- There will always be a tier that has more students and requires more buses.
- The iterative planning process focuses on that peak tier to reduce ridership or plan extra routing capacity for that tier.



Techniques to to optimize routes

1. Make adjustments in bell times
2. Change the number of buses used
3. Change the eligibility zone for bus ridership



Option #1 (RS plan D)

Open new campus 2-tier

Morning transportation – 45-minute route time.

- 34 buses transport elementary students to school with an average of 38 riders assigned to each bus.
- 31 buses transport high and middle school students to school. Bus capacity is nearly fully utilized with an average of 47 students assigned to each bus.
- Six buses transport non-public students.
- Four buses transport students with special needs.
- One bus transports students to CVCC.



Option #1 (RS plan D)

Open new campus 2-tier

Afternoon transportation – 60-minute route time.

- 31 buses transport elementary students home with an average of 43 riders assigned to each bus.
- 33 buses transport high and middle school students home with each bus averaging 44 students.
- Five buses transport non-public students.
- Four buses transport students with special needs.
- One bus transports students from CVCC.



Option # 2 (RS plan E) for the opening of the new elementary campus 3-tier option

This plan schedules elementary school first in the morning and provides bus capacity that allows flexibility for an increase in ridership without adding additional buses.

- 32 buses transport elementary school students to school with an average of 37 riders assigned to each bus.
- 23 buses transport high school students to school with an average of 25 riders.
- 25 buses transport middle students to school with an average of 34 riders.
- Six buses transport non-public students, some nonpublic schools will be shuttled but three will be on dedicated buses.
- Four buses transport students with special needs.
- One bus transports students to CVCC



Option # 2 (RS plan E) for the opening of the new elementary campus 3-tier option

This plan schedules elementary school first in the afternoon. HS dismissal bell times set 60 minutes later than the elementary.

- 28 buses transport elementary school students home, with 7 buses scheduled to exclude the high school tier, allowing for an extended elementary route.
- 21 buses transport high school students home with an average of 25 riders assigned to each bus.
- 28 buses transport middle students home with an average of 32 riders assigned to each bus.
- Four buses transport non-public students.
- Four buses transport students with special needs.
- One bus transports students from CVCC.



One Year option #3 (RS plan B)

Morning transportation – 34 buses are required.

- 30 buses transport the elementary school students.
- 21 buses transport high school students.
- 25 buses transport the middle school students.
- Six buses transport non-public students.
- Four buses transport students with special needs.
- One bus transports students to CVCC.
- 35 minute route times



One Year option #3 (RS plan B)

Afternoon transportation – 34 buses are required.

- 27 buses transport elementary school students home, four elementary buses are scheduled to omit a high school route.
- 21 buses transport high school students home.
- 25 buses transport middle school students home.
- Four buses transport non-public students.
- Four buses transport students with special needs.
- One bus transports students from CVCC.
- 60 minute elementary route time
- HS and MS 35 minute route times



The estimated cost for each option:

- 1) The current routing plan (Plan A) using 40 buses. The annual estimated cost of operation for this plan will be \$3,067,661.
- 2) Implementing Plan B, a three-tier routing plan using 34 buses, has an estimated operational cost of \$2,586,045 annually.
- 3) Implementing Plan D, a two-tier routing plan using 39 buses, has an estimated operational cost of \$2,884,664 annually.
- 4) Implementing Plan E, a three-tier routing plan using 36 buses, has an estimated operational cost of \$2,792,423 annually.



Proposal options

- The District should consider a change in the bell times for the school buildings.
- The Administration should consider engineering the bus parking arrangements that reduce the walking distance for students and prioritize bus egress to increase route time efficiency and students' safety.
- The Administration should consider adjusting bell schedules of all buildings congruently. A change in one building bell schedule without a corresponding time for the remaining buildings will have an impact on the bus route time.



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